

PADHAISHURU

GATE Preparation Platform

GATE PI

Production & Industrial Engineering

TREND-BASED MOCK PAPER

— 01 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	9 subjects

Difficulty Distribution

Easy: 17 questions (26%)

Moderate: 39 questions (60%)

Difficult: 9 questions (14%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
These mock papers are constructed using analysis of historical GATE
PYQ patterns.
Historical patterns do not guarantee future examination questions.

GENERAL INSTRUCTIONS

GATE PI — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1.

1M [no -ve]

[General Aptitude]

10% increase from 100:

*Enter your answer here***Q2.**

1M [-0.33]

[Engineering Mathematics]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
- (B) $P(A''|B)/P(B)$
- (C) $P(A)+P(B)$
- (D) $P(A)P(B)$

Q3.

1M [no -ve]

[Manufacturing Processes]

Arc welding temperature:

*Enter your answer here***Q4.**

1M [-0.33]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q5.

1M [-0.33]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q6.

1M [-0.33]

[Mechanics of Materials]

Tensile stress formula:

- (A) F/A
- (B) $F \times A$
- (C) F/A^2
- (D) A/F

Q7.

1M [-0.33]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q8.

1M [no -ve]

[Thermal Engineering]

Convection involves:

*Enter your answer here***Q9.**

1M [-0.33]

[Metrology & Inspection]

Vernier caliper least count:

- (A) 0.1 mm
- (B) 0.01 mm
- (C) 0.001 mm
- (D) 1 mm

Q10.

1M [-0.33]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q11.

1M [no -ve]

[Manufacturing Processes]

Sand mould uses:

*Enter your answer here***Q12.**

1M [no -ve]

[Manufacturing Processes]

Sand mould uses:

*Enter your answer here***Q13.**

1M [-0.33]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q14.

1M [no -ve]

[Manufacturing Processes]

Sand mould uses:

*Enter your answer here***Q15.**

1M [-0.33]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q16.

1M [-0.33]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q17.

1M [-0.33]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q18.

1M [-0.33]

[Thermal Engineering]

Convection involves:

- (A) Conduction only
- (B) Mass motion
- (C) Radiation only
- (D) None

Q19.

1M [-0.33]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q20.

1M [no -ve]

[Industrial Engineering]

CPM stands for:

*Enter your answer here***Q21.**

1M [no -ve]

[Metrology & Inspection]

Vernier caliper least count:

*Enter your answer here***Q22.**

1M [-0.33]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q23.

1M [-0.33]

[General Aptitude]

10% increase from 100:

- (A) 110
- (B) 115
- (C) 120
- (D) 105

Q24.

1M [-0.33]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q25.

1M [-0.33]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q26.

1M [-0.33]

[Engineering Mathematics]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
- (B) $P(A''|B)/P(B)$
- (C) $P(A)+P(B)$
- (D) $P(A)P(B)$

Q27.

1M [no -ve]

[General Aptitude]

10% increase from 100:

*Enter your answer here***Q28.**

1M [-0.33]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q29.

1M [no -ve]

[Metrology & Inspection]

Vernier caliper least count:

*Enter your answer here***Q30.**

1M [-0.33]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q31.

2M [no -ve]

[Operations Research]

Optimal solution of LP:

*Enter your answer here***Q32.**

2M [no -ve]

[Machine Design]

Keyway weakens shaft by:

*Enter your answer here***Q33.**

2M [no -ve]

[Manufacturing Processes]

Sand mould uses:

*Enter your answer here***Q34.**

2M [-0.66]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q35.

2M [no -ve]

[Mechanics of Materials]

Tensile stress formula:

*Enter your answer here***Q36.**

2M [-0.66]

[Metrology & Inspection]

Vernier caliper least count:

- (A) 0.1 mm
- (B) 0.01 mm
- (C) 0.001 mm
- (D) 1 mm

Q37.

2M [-0.66]

[Metrology & Inspection]

Vernier caliper least count:

- (A) 0.1 mm
- (B) 0.01 mm
- (C) 0.001 mm
- (D) 1 mm

Q38.

2M [no -ve]

[Metrology & Inspection]

Vernier caliper least count:

*Enter your answer here***Q39.**

2M [-0.66]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q40.

2M [no -ve]

[Industrial Engineering]

CPM stands for:

- (A) Critical Path Method
- (B) Critical Process Method
- (C) Critical Project Method
- (D) None

Q41.

2M [-0.66]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q42.

2M [-0.66]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q43.

2M [-0.66]

[Manufacturing Processes]

Sand mould uses:

- (A) Pattern
- (B) Core
- (C) Both
- (D) None

Q44.

2M [-0.66]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q45.

2M [-0.66]

[General Aptitude]

10% increase from 100:

- (A) 110
- (B) 115
- (C) 120
- (D) 105

Q46.

2M [-0.66]

[Thermal Engineering]

Convection involves:

- (A) Conduction only
- (B) Mass motion
- (C) Radiation only
- (D) None

Q47.

2M [no -ve]

[Manufacturing Processes]

Arc welding temperature:

Enter your answer here

Q48.

2M [-0.66]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q49.

2M [-0.66]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q50.

2M [-0.66]

[Manufacturing Processes]

Arc welding temperature:

- (A) ~3000K
- (B) ~6000K
- (C) ~10000K
- (D) ~500K

Q51.

2M [no -ve]

[Metrology & Inspection]

Vernier caliper least count:

*Enter your answer here***Q52.**

2M [-0.66]

[General Aptitude]

10% increase from 100:

- (A) 110
- (B) 115
- (C) 120
- (D) 105

Q53.

2M [-0.66]

[Engineering Mathematics]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
- (B) $P(A \cap B)/P(B)$
- (C) $P(A)+P(B)$
- (D) $P(A)P(B)$

Q54.

2M [no -ve]

[Machine Design]

Keyway weakens shaft by:

*Enter your answer here***Q55.**

2M [-0.66]

[Mechanics of Materials]

Tensile stress formula:

- (A) F/A
- (B) $F \times A$
- (C) F/A^2
- (D) A/F

Q56.

2M [-0.66]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q57.

2M [-0.66]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q58.

2M [-0.66]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q59.

2M [-0.66]

[Metrology & Inspection]

Vernier caliper least count:

- (A) 0.1 mm
- (B) 0.01 mm
- (C) 0.001 mm
- (D) 1 mm

Q60.

2M [-0.66]

[Machine Design]

Keyway weakens shaft by:

- (A) 10%
- (B) 20%
- (C) 30%
- (D) 50%

Q61.

2M [-0.66]

[Engineering Mathematics]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
- (B) $P(A''|B)/P(B)$
- (C) $P(A)+P(B)$
- (D) $P(A)P(B)$

Q62.

2M [-0.66]

[Engineering Mathematics]

Probability of event A given B = ?

- (A) $P(A)/P(B)$
- (B) $P(A''|B)/P(B)$
- (C) $P(A)+P(B)$
- (D) $P(A)P(B)$

Q63.

2M [no -ve]

[Engineering Mathematics]

Probability of event A given B = ?

*Enter your answer here***Q64.**

2M [-0.66]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible

Q65.

2M [-0.66]

[Operations Research]

Optimal solution of LP:

- (A) Unique
- (B) Multiple
- (C) Unbounded
- (D) All possible